Eur. J. Biochem. **268**, 1964–1971 (2001) © FEBS 2001

Mechanism of phosphoryl transfer by nucleoside diphosphate kinase

pH dependence and role of the active site Lys16 and Tyr56 residues

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Nucleoside diphosphate (NDP) kinase phosphorylates nucleoside diphosphates with little specificity for the base and the sugar. Although nucleotide analogues used in antiviral therapies are also metabolized to their triphosphate form by NDP kinase, their lack of the 3′-hydroxyl of the ribose, which allows them to be DNA chain terminators, severely impairs the catalytic efficiency of NDP kinase. We have analyzed the kinetics parameters of several mutant NDP kinases modified on residues (Lys16, Tyr56, Asn119) interacting with the γ-phosphate and/or the 3′-OH of the Mg^{2+}-ATP substrate. We compared the relative contributions of the active-site residues and the substrate 3′-OH for point mutations on Lys16, Tyr56 and Asn119. Analysis of additional data from pH profiles identify the ionization state of these residues in the enzyme active form. X-ray structure of K16A mutant NDP kinase shows no detectable rearrangement of the residues of the active site.

Keywords: phosphoryl transfer; substrate assisted catalysis; dideoxynucleotides; tyrosyl titration.

Nucleoside diphosphate kinase (ATP: nucleoside diphosphate phosphotransferase, EC 2.7.4.6; NDP kinase) catalyses the phosphorylation of nucleoside diphosphates into triphosphates. The reaction is ping-pong according to:

$$N_1TP + E \rightarrow E \sim P + N_1DP \quad \text{(reaction A)}$$

$$E \sim P + N_2DP \rightarrow E + N_2TP \quad \text{(reaction B)}$$

where N_1TP is usually ATP. It involves a covalent intermediate with a histidine residue phosphorylated in the active site. The enzyme shows little specificity for the base and uses ribonucleotides as well as deoxy-ribonucleotides as substrates [1].

Genes coding for NDP kinases were first identified in 1990 [2–5], and more than 70 NDP kinase genes have been cloned to date in prokaryotes and eukaryotes. The crystal structure of several NDP kinases has been determined at high resolution showing a characteristic subunit fold with a βαββαβ motif also called the ferredoxin fold [6]. The structure of the active site is highly conserved from prokaryotes to eukaryotes as could be expected from proteins whose sequences are 40% identical or more. The NDP kinase from *Dictyostelium discoideum* has often served as a reliable model to study the catalytic reaction at the atomic level, not only because it was the first eukaryotic NDP kinase to be solved but because it is particularly easy to crystallize. Generally, the active site appears as a preformed rigid template to which the nuleotide binds with very little, if any, change in protein conformation [7]. A structure of the phosphorylated enzyme where the phosphate is covalently bound to the active site His122 shows no major conformational change relative to the free protein [8]. The structure of several complexes of NDP kinases with nucleoside diphosphates [7–10] provides a precise picture of the binding mode of a nucleotide in the active site. In all cases, the base is near the protein surface while the ribose and phosphate moieties are located deeper inside the active site. Of particular interest is the 3′-OH group of the sugar, which is involved in a network of hydrogen bonds with conserved residues of the active site (Asn 119 and Lys16). The 3′-OH also donates an intramolecular hydrogen bond to O_7, stabilizing a particular conformation of the nucleotide which facilitates the phosphoryl transfer on His122 and N_1DP expulsion in reaction A. A model of substrate assisted cataysis has been proposed where the nucleotide is constrained by the preformed template of the active site [7,11].

The importance of the 3′-OH group of the ribose in the nucleotide substrate for the NDP kinase reaction is particularly well emphasized by the study of the phosphorylation of nucleoside analogs used in antiviral therapies. These drugs (AZT, d4T, 3TC etc.) are targeted to the reverse transcriptase which incorporates them in nascent viral DNA chains, thus blocking further elongation due to the lack of a 3′-OH in their ribose moiety. However, in order to be active, nucleoside analogs must be phosphorylated into

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Abbreviations AZT, 3′-deoxy-3′-azidothymidine; d4T, 2′,3′-didehydro-2′,3′-dideoxythymidine; ddATP, 2′,3′-dideoxyadenosine triphosphate; NDP, nucleoside diphosphate; Dd-NDP kinase, *Dictyostelium discoideum* nucleoside diphosphate kinase.

Enzymes: ATP: nucleoside diphosphate phosphotransferase (EC 2.7.4.6); L-lactate dehydrogenase (EC 1.1.2.7); phospho-*enol*pyruvate kinase (EC 2.7.4.0).

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Note: The coordinates of the mutant K16A NDP kinase have been deposited at the PDB with the coordinate entry 1HHQ.

(Received 25 October 2000, revised 16 January 2001, accepted 31 January 2001)

5′-triphospho-derivatives by cellular kinases and the last step in this phosphorylation pathway is catalyzed by NDP kinase. We have recently shown that NDP kinase activity is decreased by a factor 10^3-10^4 with these substrates [11–14] and by solving the structure of several binary complexes, it appears that this is mainly due to the lack of this critical 3′-OH [14,15].

In order to understand better the relative contributions of the 3′-OH of the substrate and of active-site residues interacting with the ribose, we have examined the role of Lys16, Asn119 and Tyr56 in *Dictyostelium* NDP kinase by site-directed mutagenesis. Here we present a study of the pH dependence of NDP kinase reaction catalyzed by the wild-type and the mutant proteins. This study allows us to identify the amino acids responsible for the pH-activity profile. The X-ray structure of the K16A mutant protein is described and the respective contributions of Lys16, Tyr56, Asn119 and 3′-OH of the nucleotide substrate to catalysis are discussed.

MATERIALS AND METHODS

Chemicals and site-directed mutagenesis

All nucleotides, phospho-*enol*pyruvate, lactate dehydrogenase and pyruvate kinase were purchased from Roche Molecular Biochemicals. NADH was from Sigma. The other reagents were from Merck.

Oligonucleotide-directed *in vitro* mutagenesis was performed as described by [16]. The oligonucleotide 5′-CCTTGCTGTTA**G**ACCAGACG-3′ was used to obtain the mutation K16R. The altered base as compared to the wild-type sequence is bold underlined. The mutation was verified by DNA-sequencing. Mutagenesis to obtain mutants K16A and N119A was previously described [17].

Purification of NDP kinases

Wild-type and mutated NDP kinases from *Dictyostelium* were overexpressed in *Escherishia coli* (XL1-Blue) using plasmid *pndk* as described with small modifications [2]. The cell extract was loaded at pH 8.4 onto Q-Sepharose Fast Flow which retains only *E. coli* NDP kinase. The flow-through was adsorbed on Orange A (Dyematrex, Amicon) at pH 7.5. After washing with Tris buffer, the enzyme was eluted by a linear gradient of NaCl (0–1.5 M) in 50 mM Tris/HCl pH 7.5. The protein was concentrated with an Amicon ultrafiltration cell, equilibrated in 50 mM Tris/HCl pH 7.5, and stored frozen at −20 °C. All recombinant enzymes were purified to homogeneity as judged by SDS/polyacrylamide gel electrophoresis. The concentration of wild-type NDP kinase and Lys16 mutant proteins was determined using an absorbance coefficient of $A_{280} = 0.55$ for a 1 mg·mL^{-1} solution. A value of $A_{280} = 0.50$ was used for the Y56A mutant. Enzyme concentration is expressed as the subunit concentration using $M_r = 17\ 000$.

For inactivation studies, NDP kinase was incubated at 20 °C during 24 h in 0.5–7 M urea in T buffer (50 mM Tris/HCl containing 5 mM $MgCl_2$ and 75 mM KCl). The enzyme was used at the final concentration of 0.02 mg·mL^{-1}. The residual activity was measured with the coupled enzymes described below. Low residual urea concentrations (less than 40 mM) do not interfere with the enzymatic assay.

The NDP kinase activity at the steady state was measured at 20 °C by monitoring ADP by a coupled assay using lactate dehydrogenase and pyruvate kinase as previously described [18]. In order to determine the kinetic parameters, the initial rate was measured with [ATP] and [dTDP] varying together while maintaining their concentrations at a constant ratio, usually 0.05 or 0.1. As described in [19], the double reciprocal plot is linear and the [1/v-axis] vertical intercept gives the reciprocal of the true V_{max}. The K_m for both ATP and dTDP was determined from the slope of the reciprocal plot and the ratio values.

NDP kinase activity was measured in the pH range 5.0–9.5 using the same assay where T buffer was replaced by AMT isoionic buffer (50 mM acetic acid, 50 mM Mes and 100 mM Tris/HCl) [20]. It was verified that the stability of NDP kinase and the auxiliary enzymes is unaffected by the AMT mix and that the coupled assay is a valid measurement of activity in the whole pH range.

Spectrophotometric pH titration of NDP kinases

The pH of a solution of NDP kinase (47–50 μM) in Tris/HCl 25 mM pH 7.5 containing KCl 75 mM and 5% glycerol was measured in a one-mL quartz spectrophotometric cell after addition of NaOH 1 M (or HCl 1N) in 1 μL aliquots. The absorbance at 295 nm was then measured in a Uvikon 932 spectrophotometer. The total addition of NaOH was less than 25 μL. After correction for dilution, the concentration of tyrosinate ions was calculated using a molar absorption coefficient of 2500 M^{-1}·cm^{-1} at 295 nm [21].

Crystallization of K16A NDP kinase and data collection

Crystals of the *Dictyostelium* K16A NDP kinase were grown in hanging drops containing 5 mg·mL^{-1} protein in 50 mM Tris/HCl pH 7.5, 1 M ammonium sulfate and 20 mM $MgCl_2$ over pits containing 2 M ammonium sulfate in the same buffer. Crystals appeared within two weeks. They belong to the hexagonal space group $P6_322$ with unit cell a = b = 72.15 Å, c = 107 Å. The asymetric unit contains a monomer.

Diffraction data to 2.1 Å resolution were collected at λ = 1.542 Å on a Rigaku X-ray generator with a R-axis image plate detector. Data were processed with [22], MOSFLM [23] and the CCP4 program suit. Statistics are given in Table 1. As the crystals were isomorphous to those of wild-type NDP kinase, the 1.8 Å refined model [24] could be used to calculate phases. The resulting electron density map showed the missing Lys16 side chain and little change elsewhere. Refinement with XPLOR [25] was straightforward. In the last annealing cycle, 10% of the reflections were omitted and used to calculate R_{free}. Residues 2–5 are missing in the final model, being disordered in the mutant as in the wild-type enzyme. The model contains 103 water molecules and a sulfate group interpreting residual electron density found at the active site. Refinement statistics are in Table 1.

Fluorometric binding studies and stopped-flow kinetic experiments

The affinity of dTDP for NDP kinase was determined by following the intrinsic fluorescence of the F64W mutant of *Dictyostelium* NDP kinase upon nucleotide binding as described in [13].

Table 1. Statistics on X-ray data and refinement. For resolution, A 2 σ cutoff on intensities was applied when estimating the number of unique reflections, completeness and *R* factors. $R_{merge} = \Sigma\ |I(h) - \langle I(h)\rangle| / \Sigma\ I(h)$. $R_{cryst} = \Sigma||Fo| - |Fc||/\Sigma|Fo|$. R_{free} is the same ratio calculated on 10% of the total reflections excluded from the last cycle of annealing refinement. Values in parentheses are for the outer shell of resolution (2.2–2.1 Å). Bond length rmsd is the root-mean-square discrepancy from library values. The coordinates have been deposited to the Protein Data Bank (file name 1HHQ).

Space group	$P6_322$
Asymmetric unit	monomer
Cell parameters: a = b, c (Å)	72.2, 107.0
Resolution (Å)	2.1
Unique reflections	11016
Redundancy	4.3
Completeness (%)	98.2
R_{merge} (%)	5.0
R_{cryst} (%)	20.1 (25.9)
R_{free} (%)	24.8 (32.1)
Bond length rmsd (Å)	0.013
Bond angle rmsd (°)	1.91
Torsion angle rmsd (°)	1.37

Stopped-flow kinetic experiments were performed with a Hi-Tech DX2 microvolume stopped-flow reaction analyzer equipped with a high-intensity xenon lamp as described [14]. The excitation wavelength was 304 nm, with a 2-mm excitation slit and a 320-nm cutoff filter at the emission. Mixing was achieved in less than 2 ms. After mixing NDP kinase (1 μM) and ATP or ddATP (10–500 μM), the intrinsic protein fluorescence was recorded for 1–200 s. In each experiment 400 pairs of data were recorded. The data from three to four identical experiments were averaged and fitted to a number of nonlinear analytical equations using the software provided by Hi-Tech. All curves fitted to single exponentials.

As previously described [13], the data were analyzed using a two-step mechanism with a fast nucleotide binding step followed by a slow phosphoryl transfer. For 'poor' substrates such as dideoxynucleotide triphosphates, the observed single step could be attributed to the phosphotransfer between the nucleotide and the enzyme. The same model was used here for ATP as k_{cat}/K_m^{ATP} measured at the steady state amounts to the same value as the catalytic efficiency measured in stopped flow kinetic experiments. Saturation could not be obtained with the concentrations of ATP or ddATP used here. Therefore, we measured the catalytic efficiency of the enzyme phosphorylation k_{SF} which is equivalent to a second order constant.

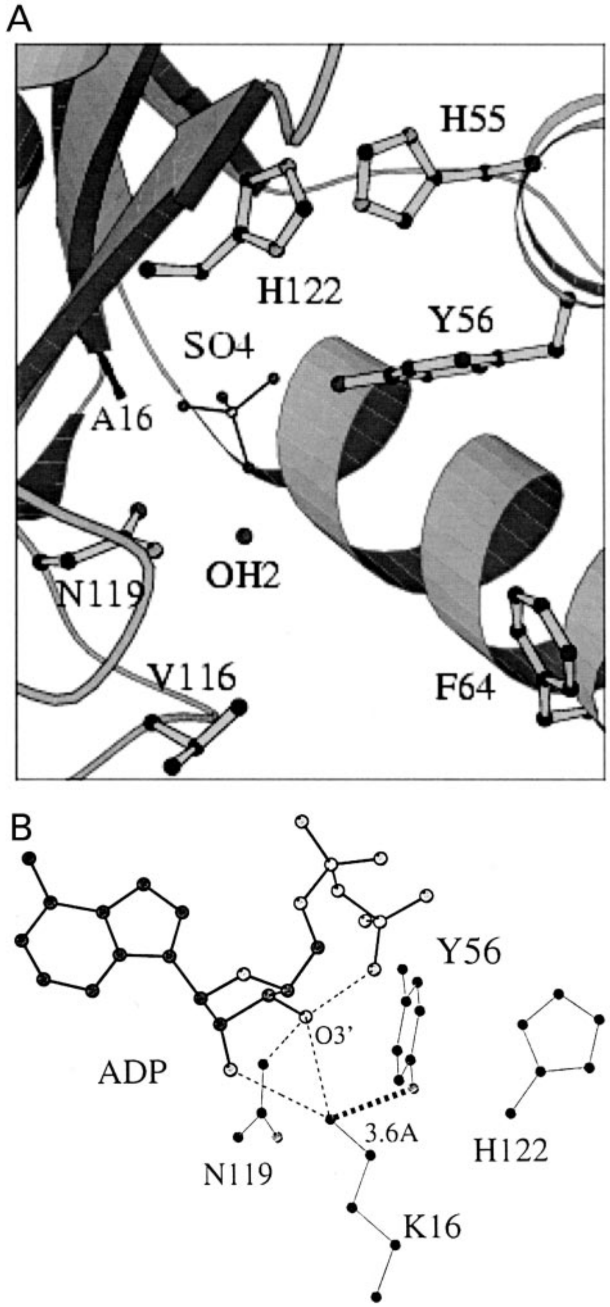

Fig. 1. Stucture of the active site of mutant and wild-type NDP kinase. (A) Structure of the active site of K16A mutant NDP kinase from *Dictyostelium* (1HHQ.pdb coordinates). (B) Active site of NDP kinase from *Dictyostelium* with bound ADP-AlF_3 showing the pyrophosphate moiety of ADP, the bound Mg^{2+} and surrounding protein side chains. Dash lines indicate the atoms at H-bond distances (1KDN.pdb coordinates).

RESULTS AND DISCUSSION

Structural properties of K16A, K16R and Y56A mutant NDP kinases

The mutant NDP kinases (K16A, K16R, Y56A, N119A) were expressed at high level as recombinant proteins in *E. coli* and were purified to homogeneity as described in Materials and methods. Their UV and fluorescence spectra were characteristic of properly folded proteins. In the case of K16A mutant protein, we obtained diffracting crystals. The structure was solved at 2.1 Å resolution (Table 1). The protein is a hexamer with D3 symmetry and a superposition of the main chain atom positions in the mutant and wild-type yields a RMS discrepancy of 0.3 Å. Very little change is observed at the active site except for the absence of the Lys16 side chain, which is partly replaced with an ordered water molecule (Fig. 1A). No main chain movement takes place, even at the mutation site, and surrounding side chains keep their conformation. A sulfate ion is seen at the active site. It is also present in ammonium sulfate grown crystals of the wild-type, and approximately occupies the position of the phosphate groups in a nucleotide substrate. Previously, we reported the structure of the N119A mutant protein, which also was very similar to that of the wild-type protein, except for a local change of the main chain conformation at the mutation site [15].

We have studied the stability of the K16R NDP kinase. The wild-type enzyme was known as stable and active till 5 M urea and was found to dissociate and inactivate concomitantly with an EC_{50} at 5.5 M urea [18]. The K16R mutant was found slightly less stable with half-inactivation at 4 M urea (data not shown). A similar decrease in urea or heat stability was previously observed for K16A and N119A mutant proteins [17]. The lower stability of the mutants may be related to the network of hydrogen bonds made by Lys16 as the amino group of the Lys16 interacts both with the phenol hydroxyl of Tyr56 and with the amide oxygen of Asn119 (Fig. 1B).

Kinetic parameters of mutant NDP kinases with natural substrates and with antiviral analogs

The kinetic parameters at the steady state of the four mutants analyzed in this study were determined using ATP and dTDP as substrates (Table 2). None of the mutations result into an inactive enzyme. The K_m for both substrates are little affected by the mutations with a variation of less than fivefold. In contrast, for the catalytic constant (k_{cat}), a larger variation is observed in the case of Lys16 mutants, with a decrease in k_{cat} by a factor of 100 for mutant K16A. When the positive charge is restored in mutant K16R, the k_{cat} increases by a factor of four relative to the K16A mutant, showing the importance for catalysis of a positive charge at this position. The two other mutants are less affected: the activity of the Y56A protein is decreased only four fold and N119A mutant has the same k_{cat} as the wild-type (see below). In a preliminary characterization of mutants K16A, N119A and Y56A, we had measured the activities by a single assay ([ATP] = 1 mM, [dTDP] = 0.2 mM) [17]. Under these so-called 'standard conditions' the activity of K16A and Y56A mutant proteins were in the same range as the activities measured more precisely here, although they were slightly underestimated, possibly because the enzyme preparations were not pure or partly inactivated. In contrast, mutant N119A, which now shows an activity similar to the wild-type protein, was less than 1% active in the preliminary characterization. As this mutant is significantly less stable towards heat denaturation than the others [17] it is likely that the enzyme preparation was denaturated to a large extent.

A relevant parameter for a ping-pong enzyme is its catalytic efficiency (k_{cat}/K_m). When this parameter is taken into consideration (Table 2) it appears even more clearly that the mutants fall into two categories. N119A and Y56A are not or little affected (only a factor of four Y56A). In contrast K16A and K16R NDP kinases have catalytic efficiencies decreased by factors of 100 and 10, respectively. The importance of the electric charge of Lys16 in catalysis can be interpreted taking into account the numerous structural data available with NDP kinase. It is likely due to a role in neutralizing the negative charge on O_7 of ADP, which develops upon the transfer of the γ-phosphate onto His122 [26,27].

The rates of phosphoryl transfer to the protein can be measured in fast mixing experiments by monitoring the change in NDP kinase intrinsic fluorescence that occurs upon phosphorylation of the catalytic histidine [28]. This rate of phosphorylation, with mutant K16R protein using ATP as a donor, is decreased by a factor of 10 ($k_{SF}^{ATP} = 3.10^5\ \mathrm{M}^{-1}\cdot\mathrm{s}^{-1}$ as compared to $k_{SF}^{ATP} = 3.10^6\ \mathrm{M}^{-1}\cdot\mathrm{s}^{-1}$ for the wild-type), in full agreement with activity values of k_{cat}/K_m shown in Table 2. This indicates that in the mutant protein as in the wild-type protein, the phosphotransfer is the limiting step in the catalytic reaction [13]. Note that the fluorescence of the K16A protein is not quenched upon phosphorylation, an observation for which we have no explanation.

We have previously shown that NDP kinase has greatly decreased activity with ddATP as compared to ATP. Using this substrate, fluorescence stopped-flow experiments yield monoexponential curves which allows the calculatation a second order rate constant (equivalent to the catalytic efficiency) [13]. When this type of experiment is performed with the K16R mutant and ddATP, a value for phosphorylation of $k_{SF}^{ddATP} = 160\ \mathrm{M}^{-1}\cdot\mathrm{s}^{-1}$ is found (Table 2) showing a 2000-fold decrease in catalytic efficiency as compared to

Table 2. Kinetic parameters for wild-type and mutant NDP kinases at pH = 7.5. The specific activities were determined as usual in standard substrate conditions [ATP] = 1 mM, [dTDP] = 0.2 mM using the coupled assay at 20 °C and pH 7.5. One unit of enzyme activity is defined as the amount of enzyme that catalyzes the phosphorylation of 1 μmol of dTDP per minute under the standard conditions. The kinetic parameters of the enzymes were measured as a function of [ATP] with a constant ratio [dTDP]/[ATP] (ratio = 0.1 and 0.05), as recommended in [19], using the coupled assay at 20 °C and pH 7.5: k_{cat} values were from the V_{max} values and expressed for one active site, i.e. per subunit (M_r = 17 000). Values are means of 2–4 independent measurements. k_{SF}^{ddATP} represented the catalytic efficiency of the various enzymes in stopped flow kinetic experiments.

NDP kinase	Specific activity (U mg^{-1})	k_{cat} (s^{-1}) (% of wild-type)	K_m^{ATP} (mM)	K_m^{dTDP} (mM)	k_{cat}/K_m^{ATP} ($\mathrm{M}^{-1}\cdot\mathrm{s}^{-1}$)	k_{SF}^{ddATP} ($\mathrm{M}^{-1}\cdot\mathrm{s}^{-1}$)
Wild-type	2000	680 (100)	0.44	0.11	$2.2.10^6$	1300[b]
K16A	30	10 (1.5)	0.60	0.02	$1.6.10^4$	–
K16R	150	40 (6)	0.15	0.07	$2.6.10^5$	160
N119A	2000[a]	700[a] (100)[b]	0.5[a]	0.1[a]	2.10^{6}[a]	1000
Y56A	500	170 (25)	0.17	0.06	10^6	

[a] Values from [15]; [b] values from [13].

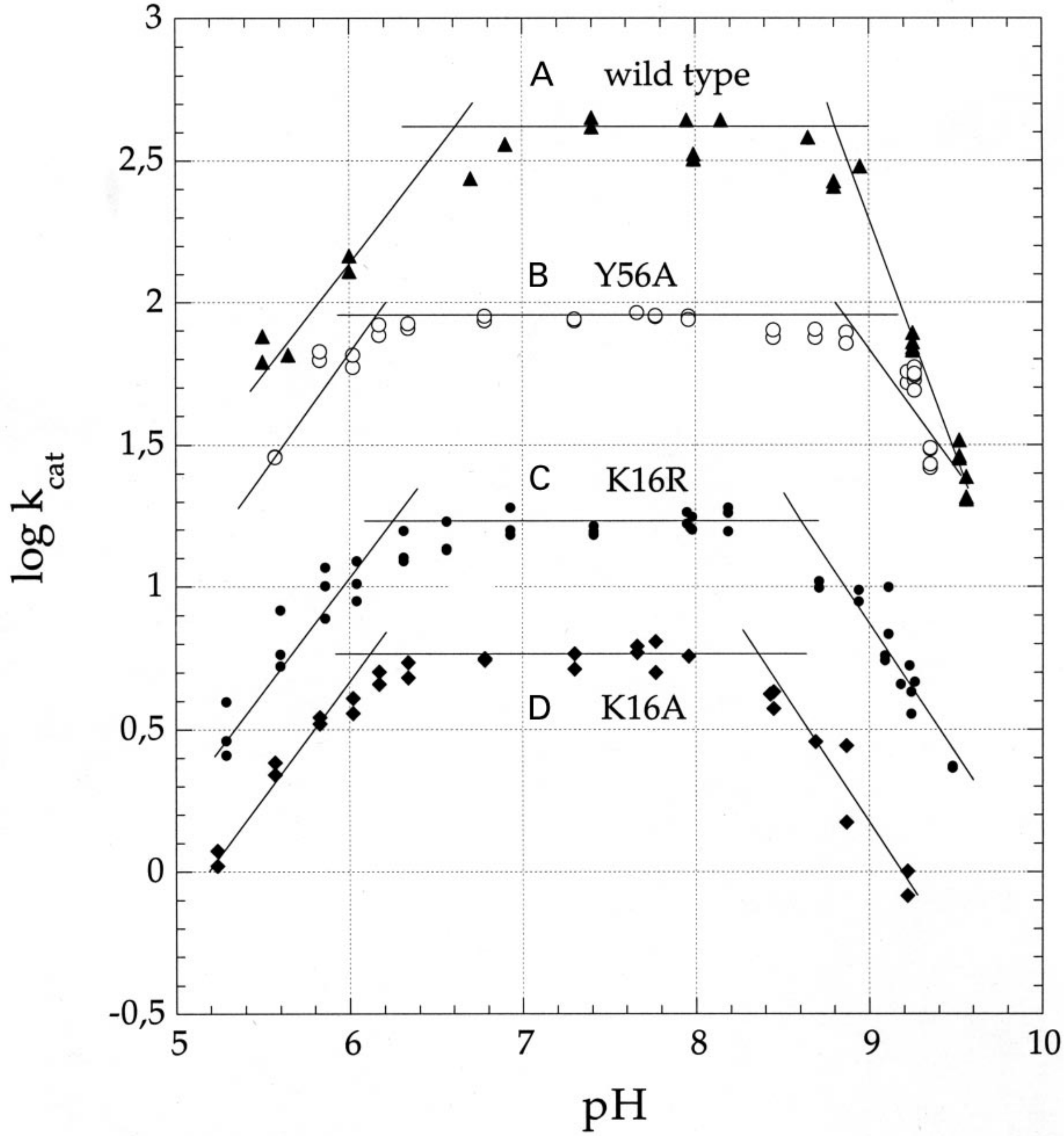

Fig. 2. pH dependence of the velocity of NDP kinase. (A) (▲) wild-type (B) (○) Y56A, (C) (●) K16R and (D) (◆) K16A. NDP kinase activity is measured in AMT buffer in 'standard' substrate conditions ([ATP] = 1 mM, [dTDP] = 0.2 mM). The measured rates correspond to the value of k_{cat} multiplied by a factor of 0.7.

the natural substrate ATP. The same decrease was found for the wild-type enzyme with ddATP as a substrate compared to ATP. In summary, the weak activity of K16R mutant with ddATP, when compared to the wild-type enzyme with ATP, comes from two additive contributions: a catalytic defect in K16R (1/10 ×) and the lack of 3′OH (1/2000 ×) of the substrate.

PH dependence of the phosphoryl transfer reaction by wild-type NDP kinase

In order to better understand the role of Lys16 and Tyr56 in the catalytic mechanism, the effect of pH on the kinetic parameters of K16A, K16R and Y56A mutant NDP kinases was studied in comparison with the wild-type (Fig. 2). In standard conditions, the rate of the reaction is strongly pH dependent and follows a bell shaped curve. The plot of log k_{cat} against pH shows three linear portions (Fig. 2, curve A) with an optimum rate between pH 7 and pH 8.5. Below pH 7 and above pH 8.5, the plot of log*V* against pH is linear with positive and negative slopes, respectively. Extrapolation of the straight lines corresponding to these slopes indicates that the ionization state of at least two residues occurs with pK_a of 6.6 and 8.8, respectively (Fig. 2).

No contribution from the nucleotide substrates are expected in the pH range 6–10: it was previously shown by ^{31}P-NMR that the pK_a of GTP γ-phosphate when bound

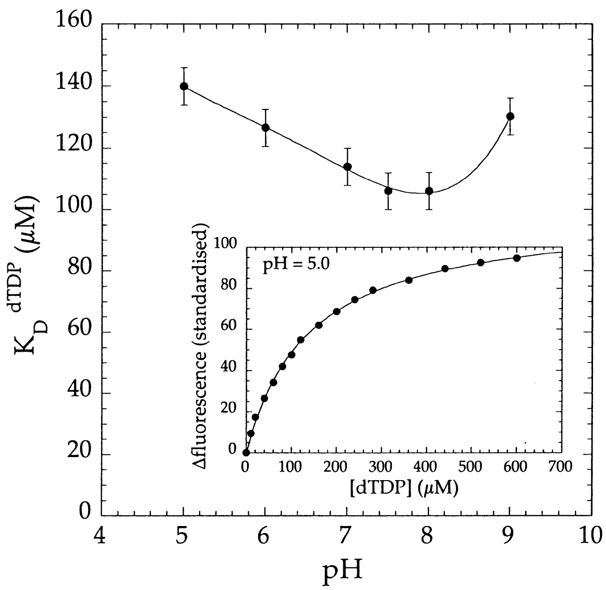

Fig. 3. pH dependence of the equilibrium constant K_d for dTDP measured by fluorescence titrations. The error bars are within 6%. Inset: the fluorescence of the F64W mutant NDP kinase in AMT buffer was measured at 330 nm with excitation at 310 nm (2 nm excitation slit and 4 nm emission slit). Successive aliquots of nucleotide were added to a 1 μM enzyme solution. The inner filter effect was negligible. Experimental binding curves were fitted to a quadratic equation for ligand-protein curve after correction for dilution.

to a protein (p21Ras) in the presence of magnesium ions is in the range pH = 3 [29]. $pK_{a1} = 6.6$ is likely to correspond to the Nδ of the catalytic His122 as this nitrogen is deprotonated to allow phosphorylation rather than to Nε of His122 which forms a hydrogen bond with Glu133 [30,31]. However, it is difficult to determine if the observed pK_a is due to the unphosphorylated or the phosphorylated form of histidine 122. This has been measured in only one exemple to our knowledge: the active site histidine has a pK_a of 7.3 when phosphorylated, and 6.3 when unphosphorylated, in the case of the N-terminal domain of enzyme I of the *E. coli* phospho-*enol*pyruvate: sugar phosphotransferase system [32]. The group corresponding to the basic pK_a is more difficult to identify as it could correspond to either Lys16 or Tyr56. These side-chains usually titrate in the pH range 9–11. In fact, given the pH dependance of the mutants, it is likely that both of these residues contribute to the observed $pK_a = 8.8$ (see below).

The K_m for both substrates (ATP and dTDP) are found to be similar at pH 6, 7.5 or 8.5 in steady state experiments (not shown). However, the K_m combines several kinetic constants and does not strictly represent substrate affinity, in particular for a ping-pong enzyme such as NDP kinase. Previously, we have designed a mutant NDP kinase (F64W), which allows the measurement of the true affinity for a nucleoside diphosphate [13]. The substitution of a Phe stacking to the nucleobase at the surface of the mutant protein into Trp has no effect on enzymatic activity and allows a direct measure of the binding of a NDP by the increase in protein fluorescence (Fig. 3, inset). Figure 3 shows the binding curve of dTDP to NDP kinase obtained by this method. The dissociation constant is $K_d = 100$ μM between pH 7 and pH 8. A 30% increase in K_d, i.e. a slight drop in affinity, is seen at $5 < pH < 7$ and at $pH > 8$.

pH dependence of the kinetic parameters of K16A, K16R and Y56A NDP kinases

Figure 2 also shows the pH dependence of the activity of the various mutant proteins measured under steady-state conditions. All mutants present a similar bell-shaped curve with a decrease at $pH > 8.5$ (Fig. 2 and Table 3). Therefore the lack of either Lys16 or Tyr56 does not prevent the decrease in activity at $pH > 8.5$. In addition, no significant change of the K_m^{ATP} or K_m^{dTDP} for K16R mutant is found for pH 6.5 or 9.0 (data not shown). In the case where the deprotonation of only one residue (either Lys16 or Tyr56) was responsible for the low activity above pH 8.5, both K16A or Y56A mutant enzymes would mimick the unprotonated enzyme, i.e. with Lys-NH_2 or Tyr-O^- and therefore would show no pH dependence at basic pH. The bell-shaped curve obtained with the mutant proteins indicate that at basic pH, more than one side chain is protonated. This is further confirmed by the difference in the slope of the curve at basic pH. The slope of $\log V$ against pH at $pH > 8.5$ shifts from -2 for the wild-type enzyme to a value around -1 for all mutants. No variation in the $+1$ slope observed for $pH < 7$ in the wild-type protein is found in the mutants, coherent with the basic nature of the residues changed by mutation (Fig. 2). A slope of -2 presumably indicates that two residues lose a proton with a similar $pK_a = 8.8$. Thus the decreased activity of Y56A NDP kinase observed at $pH > 8.5$ is likely due to Lys16 losing its proton with a $pK_a = 8.8$. Conversely for mutants K16A and K16R, the decrease of activity is due to the titration of Tyr56 with a pK_a slightly more acidic (8.3 and 8.6, respectively).

Spectrophotometric titration of Tyr residues in NDP kinases

In order to check the hypothesis that the Tyr56 side chain has a $pK_a = 8.3$ in mutant K16A, tyrosine residues were titrated by monitoring the absorbance at 295 nm between pH 7 and pH 12. No absorbance change is observed at $pH < 9.5$ for the wild-type enzyme and for the Y56A mutant (Fig. 4). The absorbance of the K16A mutant increases above pH 8 following a titration curve. Above pH 10, the enzyme denaturates rapidly, in accordance with the weak stability of the protein, and the light scattering of the solution prevents accurate spectrophotometric titrations due to protein precipitation (Fig. 4). The observed increase in A_{295} is due to the titration of one tyrosyl with a

Table 3. pK_a values of critical residues in NDP kinase wild-type and mutants. Same data as in Fig. 2.

NDP kinase	pK_a1	pK_a2
Wild-type	6.6	8.8
Y56A	6.2	8.9
K16R	6.2	8.6
K16R	6.1	8.35

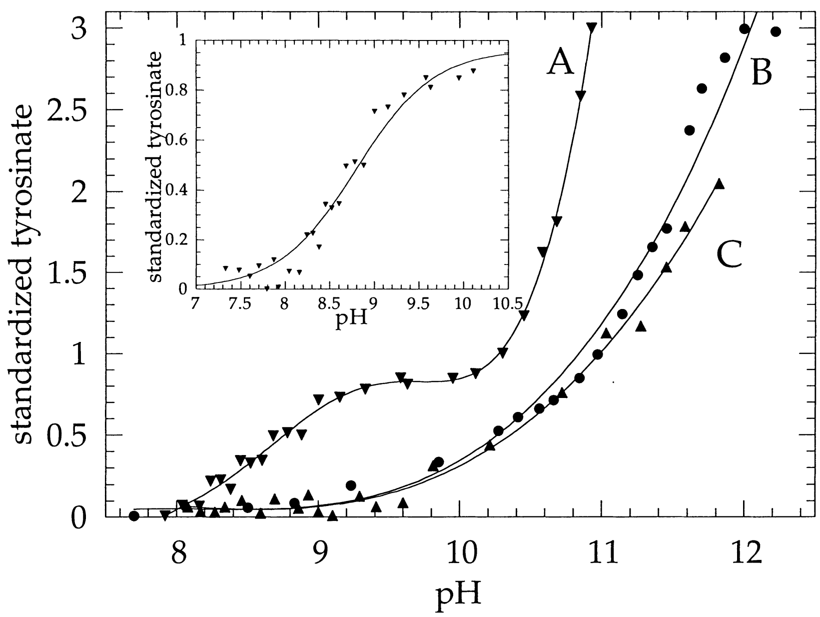

Fig. 4. Spectroscopic tyrosine titration of NDP kinases. The absorbance at 295 nm of enzyme solutions in 1 mL cell were measured as a function of pH and standardized per mole of enzyme subunit as described in Materials and methods. pH titration curves for tyrosines of K16A NDP kinase (▼) curve A; wild-type enzyme (●) curve B and K16R NDP kinase (▲) curve C. Inset: data from curve A after substracting the estimated contributions of scattered light and buried Tyr. The line represents a fit to the Henderson–Hasselbach equation with a $pK_a = 8.67$.

$pK_a = 8.3$. As a control, tyrosyl titrations were performed with the urea-unfolded wild-type NDP kinase and K16A mutant which both have three Tyr residues with a $pK_a = 10.5$. In contrast, only two Tyr residues were titrated in mutant Y56A (results not shown).

As the same pK_a of 8.3 is found both for tyrosine titration and for enzymatic activity in the K16A mutant at basic pH, we propose that this corresponds to the pK_a of Tyr56. A $pK_a = 8.3$ is not found for the wild-type enzyme presumably because Lys16 stabilizes the protonated form of Tyr56. It is unlikely that the two other tyrosine residues of NDP kinase could titrate with such a low pK_a as only Tyr56 is close to Lys16 in the enzyme and is susceptible to be affected by the K16A mutation. Therefore Lys16 appears essential to maintain Tyr56 in a protonated state which is important for NDP kinase activity. Tyr56 has been shown to bind to the phosphoryl on His122 [8] and this is presumably prevented by electrostatic repulsion if it carries a negative charge.

CONCLUSIONS

A number of crystal structures provide a background to the biochemical studies of NDP kinase reported here. They represent the wild-type enzyme with and without bound nucleotides, and several point mutants. Crucial steps in the catalytic mechanism are illustrated by the crystal structure of the phosphorylated protein [8] and of ternary complexes with ADP and either beryllium or aluminium fluoride [27]. In the complex with ADP and BeF_3, BeF_3 binds between the β-phosphate of ADP and the His122 imidazole. Its location and tetrahedral geometry reproduce those of the γ-phosphate of ATP in the Michaelis complex. AlF_3 binds at the same location, but it forms a trigonal bipyramid, which is the geometry of the γ-phosphate undergoing transfer. Thus, the complex with ADP and AlF_3 mimicks the transition state of the phosphorylation reaction.

These structural data suggest possible roles for active site groups. For instance, Tyr56, but not Lys16, interacts with the phosphohistidine in the structure of the phosphorylated protein. Conversely, Lys16 but not Tyr56, interacts with the fluoride ions mimicking oxygens on the γ-phosphate in the complexes with ADP and BeF_3 or AlF_3. We find here that Lys16 is much more important than Tyr56 in catalysis, as its substitution to Ala causes a drop in k_{cat} by two orders of magnitude instead of just a factor of four for Y56A mutant protein. Remarkably, the substitution of Asn119 has essentially no effect on catalysis, even though its side chain interacts with the 3′-OH of the sugar as well as with Lys16. In any case, none of these substitutions abolishes activity confirming that none of the active site residues is essential except for His122. The residual activity of K16 mutant proteins makes it very unlikely that Lys16 is a catalytic base as, for instance, the catalytic histidine in serine proteases. Indeed, in these enzymes, His to Ala substitutions reduce the activity by 5 or 6 orders of magnitude [33].

Our results confirm that the 3′-OH group carried by the nucleotide substrate is much more important for the NDP kinase reaction mechanism than any of the protein side chains studied here: its deletion in nucleoside analogs decreases the activity by three or four orders of magnitude. NDP kinase thus belongs to the category of enzymes where catalysis is substrate-assisted, i.e. where the substrate provides a functional group. An example are the histidine mutants of serine proteases, which are inactive on normal substrates as just mentioned, but efficiently catalyse hydrolysis of histidine-containing peptides: the imidazole group carried by the substrate takes over the function of the catalytic base [34].

This substrate-assisted catalysis in NDP kinase, may suggest a strategy for enhancing enzyme substrate specificity. A major difficulty in the use of the dideoxynucleosides analogs in antiviral therapies is the requirement for a substitution of the 3′-OH position of the sugar. Such an analog must lack the 3′-OH group of the ribose moiety to be a chain terminator and block DNA elongation by reverse transcriptase. But the same modification decreases its ability to be phosphorylated as a triphosphate derivative by NDP kinase. A compound that would conciliate the two requirements should be an excellent molecule for therapy. The strategy for enhancing NDP kinase efficiency is to choose substrates which bring their own activating group.

ACKNOWLEDGEMENTS

This work was supported by grants from Agence nationale de la Recherche contre le SIDA, Association pour la recherche sur le Cancer, and SIDACTION We acknowledge help from staff of the LURE-DCI synchrotron radiation center (Orsay, France) in data collection.

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